ALL MEMBERS

Atomic Structure — Detailed Teach-the-Teacher Script

10-member group script. Say it in your own words, but make sure every point is covered.

Each member ends with a handoff line so the flow never breaks.

Opening (All Members Together)

ALL MEMBERS

Respected teacher and friends, today our group will teach **Atomic Structure**. We will explain the Thomson model, the Rutherford model, and the limitations of classical models. We will go step by step: how the electron was discovered, how Thomson used it to build a model, how Rutherford's experiment broke that model, and finally why even Rutherford's model had to be replaced.

Member 1 — Introduction to Atomic Structure

MEMBER 01

Let me begin. Our flow today is: introduction to atomic structure, discovery of the electron, Thomson's plum pudding model and its limitations, Rutherford's gold foil experiment and its observations, Rutherford's nuclear model, a comparison of the two models, limitations of classical models, and finally the conclusion with modern atomic theory.

Member 1 — Introduction to Atomic Structure

MEMBER 01

Now the basics. The word *atom* comes from the Greek word *atomos*, which means **indivisible**. The idea is very old, but in 1808 John Dalton turned it into a scientific theory. He proposed that matter is made of tiny particles called atoms, that atoms of one element are identical, and that atoms combine in fixed ratios to form compounds. In Dalton's picture, the atom was like a tiny solid ball that can never be broken.

Member 1 — Introduction to Atomic Structure

MEMBER 01

But later experiments, such as electrolysis and electric discharge through gases, showed that atoms are connected to electricity, which means they must have smaller parts inside. So atoms are **divisible**. We now know that an atom contains three subatomic particles. The **electron** has a negative charge of 1.6×10^{-19} coulomb and a tiny mass of about 9.1×10^{-31} kg. The **proton** has the same amount of charge but positive, and it is about 1836 times heavier than the electron. The **neutron**, discovered later by Chadwick in 1932, has no charge and a mass almost equal to the proton.

Member 1 — Introduction to Atomic Structure

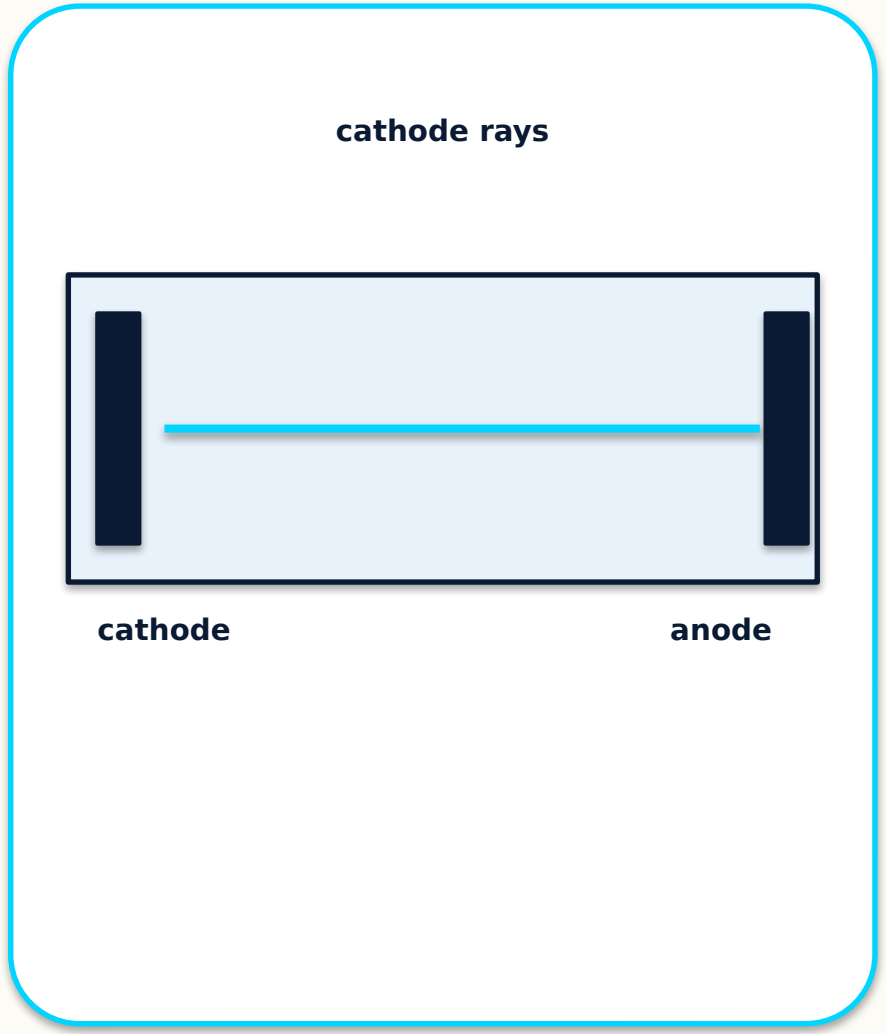
MEMBER 01

Once we know these particles exist, the big question is: how are they arranged inside the atom? Atomic models are scientists' attempts to answer exactly this question. The first step was the discovery of the electron, which my fellow member will explain.

Member 2 — Discovery of Electron

MEMBER 02

The electron was discovered by J.J. Thomson in **1897** while studying **cathode rays**. Let me explain the apparatus first. A cathode ray tube is a sealed glass tube containing a gas at **very low pressure**. When a high voltage is applied across two electrodes, the cathode and the anode, rays travel from the **cathode to the anode**. We call them cathode rays.



Member 2 — Discovery of Electron

MEMBER 02

These rays have interesting properties. They travel in straight lines, they cast the shadow of any object placed in their path, they make a small paddle wheel rotate, and they make the glass or a fluorescent screen glow. Most importantly, they are **deflected by electric and magnetic fields**. In an electric field they bend towards the positive plate, which proves that cathode rays are **negatively charged**.

Member 2 — Discovery of Electron

MEMBER 02

Thomson then adjusted the electric and magnetic fields so that their deflections cancelled each other. Using this, he measured the **charge-to-mass ratio, e/m = 1.76 × 10^{11} C/kg**. This value is about 1800 times larger than the value for a hydrogen ion, which means the particle is extremely light. And the ratio stayed the same whatever gas or cathode material he used. So these particles are a common part of every atom. These particles were named **electrons**.

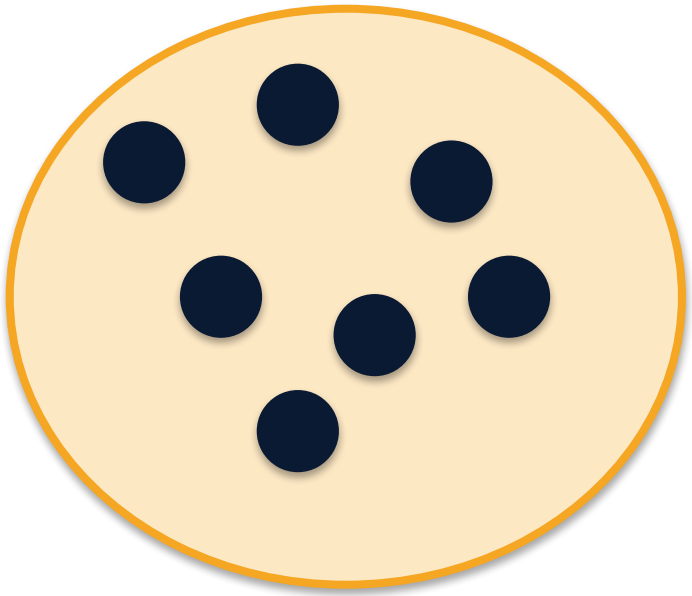
The conclusion is simple: the atom is **divisible**. Later, Millikan's oil drop experiment measured the charge of the electron separately. Now my fellow member will explain the model Thomson built using this discovery.

Member 3 — Thomson Plum Pudding Model

MEMBER 03

Once Thomson knew that electrons exist inside atoms, he had to answer: if electrons are negative and the atom is neutral, where is the positive charge? In **1904** he proposed his model.

According to Thomson, the atom is a **sphere of positive charge**, with a radius of about 10^{-10} metre. The **electrons are embedded inside** this sphere. That is why it is called the **plum pudding model**: the pudding is the positive charge and the plums are the electrons. It is also called the **watermelon model**, where the red pulp is the positive charge and the seeds are the electrons.



positive charge

electrons

Member 3 — Thomson Plum Pudding Model

MEMBER 03

The key idea is that the **total positive charge equals the total negative charge**, so the atom is **electrically neutral**. The positive charge is **spread uniformly throughout the atom**, and there is **no nucleus** in this model. Thomson also said electrons can vibrate about their positions, and he used this to try to explain how atoms emit light.

The model could explain neutrality and the fact that electrons can be removed from atoms. But it soon ran into serious trouble, which my fellow member will now explain.

Member 4 — Limitations of Thomson Model

MEMBER 04

Thomson's model failed for several reasons.

First, it could not explain large-angle alpha scattering. If positive charge is spread over the whole atom, the electric field inside is weak, so an alpha particle should only be pushed slightly. Also, an alpha particle is about 7300 times heavier than an electron, so electrons cannot deflect it either. It is like a bus hitting ping-pong balls: the bus will not bounce back. The model predicted only tiny deflections, but experiments showed otherwise.

Second, it did not introduce a nucleus. It assumed that positive charge is spread throughout the atom.

Member 4 — Limitations of Thomson Model

MEMBER 04

Third, it could not properly explain line spectra. Real atoms emit light only at particular wavelengths, but Thomson's vibrating electrons could not give these exact lines.

Fourth, it could not fully explain atomic stability. A set of charges sitting still cannot stay in stable balance using electrostatic forces alone.

Finally, the model **failed to match Rutherford's experiment**. So Rutherford proposed a new model. Let us see how he tested Thomson's idea.

Member 5 — Rutherford Gold Foil Experiment (Setup)

MEMBER 05

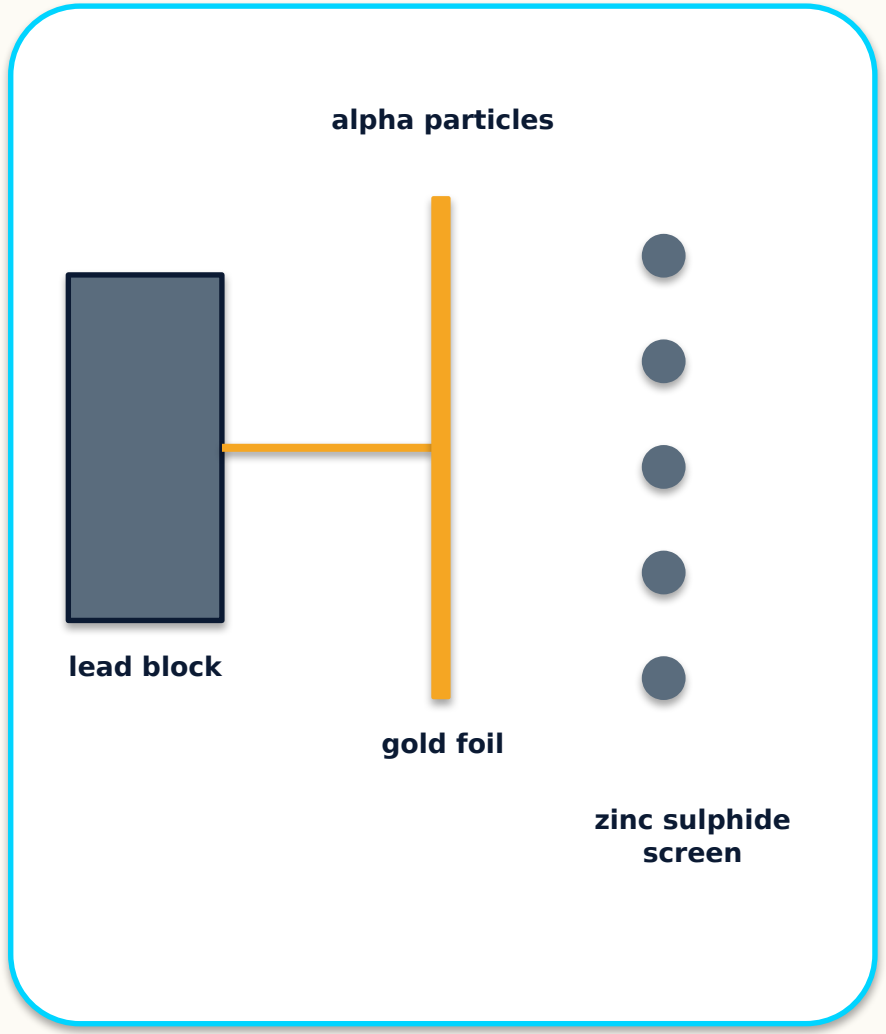
This experiment was performed by **Rutherford, Geiger, and Marsden** around 1909 at Manchester. The **aim** was to test Thomson's model by firing fast particles at atoms and watching how they get deflected.

Now the setup. The **alpha source** was a radioactive material like **radium or polonium**. Alpha particles are helium nuclei, carrying a charge of +2e and moving at very high speed, about 10^7 metre per second. The source was placed inside a **lead block** with a narrow hole, which produced a thin, narrow beam of alpha particles, because lead absorbs the rest.

Member 5 — Rutherford Gold Foil Experiment (Setup)

MEMBER 05

The beam was aimed at a **very thin gold foil**, only a few hundred atoms thick. Gold was chosen because it is highly malleable and can be hammered into extremely thin sheets, and its nucleus has a large charge of +79. Around the foil was a **zinc sulphide screen**. When an alpha particle hits it, it produces a tiny flash of light called a **scintillation**. A **microscope** was used to observe and count these scintillations at different angles. The whole apparatus was kept in a **vacuum** so that alpha particles would not scatter off air molecules.



Member 5 — Rutherford Gold Foil Experiment (Setup)

MEMBER 05

Expected result: according to Thomson's model, almost all alpha particles should go through with only **small deflections**. But the actual results were very surprising, as my fellow member will explain.

Member 6 — Observations of Rutherford Experiment

MEMBER 06

There were three main observations, and each gave a conclusion.

Observation 1: Most alpha particles passed straight through the foil without any deflection. **Conclusion:** the atom is **mostly empty space**.

Observation 2: Some alpha particles were deflected by small angles. **Conclusion: positive charge exists inside the atom**, and it repelled the positively charged alpha particles. About 0.14% of particles were deflected by more than 1 degree.

Member 6 — Observations of Rutherford Experiment

MEMBER 06

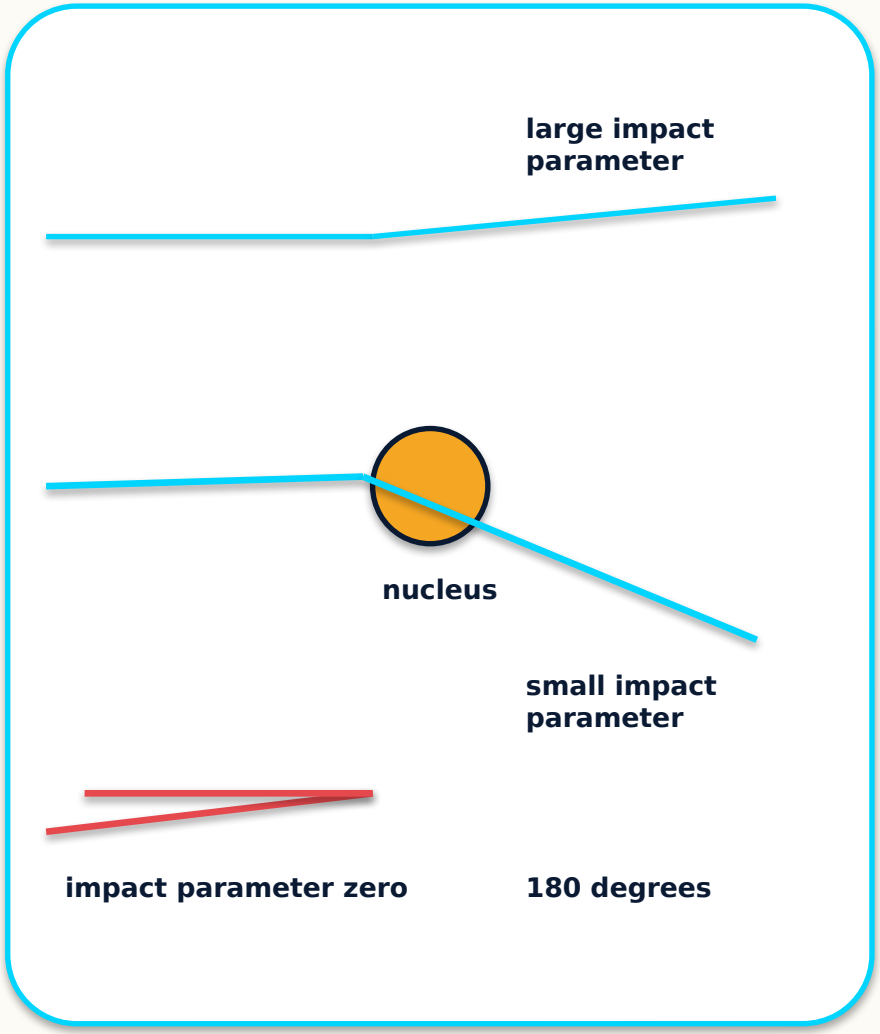
Observation 3: Very few alpha particles, about 1 in 8000, were deflected through large angles of more than 90 degrees, and some even bounced straight back. **Conclusion:** the positive charge and almost all the mass are concentrated in a **tiny, dense nucleus**. Only a head-on collision with such a concentrated charge could throw back a fast, heavy alpha particle. Rutherford later said it was like firing a huge shell at a piece of tissue paper and seeing it come back. This was the most surprising observation.

Member 6 — Observations of Rutherford Experiment

MEMBER 06

The amount of deflection depends on the **impact parameter**, which is the perpendicular distance between the alpha particle's original path and the centre of the nucleus. A **small impact parameter** means a close approach and a large deflection, and a head-on approach, with impact parameter zero, makes the particle bounce back at 180 degrees. A **large impact parameter** means a small deflection.

From this data, Rutherford built a new model, which my fellow member will explain.



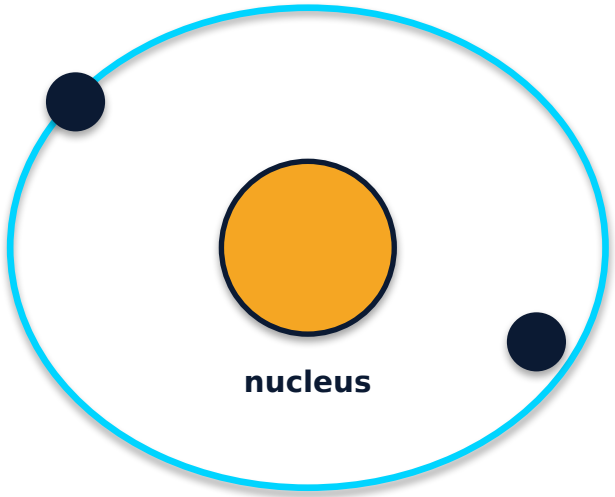
Member 7 — Rutherford Nuclear Model

MEMBER 07

Based on these observations Rutherford proposed the **nuclear model** in 1911.

According to it, the atom has a **small, dense, positively charged nucleus** at the centre. **Most of the atom's mass is present in the nucleus.** The **electrons revolve around the nucleus** in circular orbits, and **most of the atom is empty space**. Because electrons orbit the nucleus like planets orbit the sun, it is also called the **planetary model**.

What keeps electrons in orbit? Just as gravity keeps planets in orbit, the **electrostatic force of attraction** between the positive nucleus and the negative electron provides the **centripetal force**. We can write this as: the Coulomb force equals mv^2/r.



Member 7 — Rutherford Nuclear Model

MEMBER 07

The sizes are important. The **nucleus radius is about 10^{-15} to 10^{-14} metre**, while the **atom radius is about 10^{-10} metre**. That means the atom is about 10,000 to 100,000 times larger than its nucleus. If the atom were as big as a football stadium, the nucleus would be like a tiny seed at the centre.

This model explains alpha scattering perfectly: the nucleus has a strong electric field close to it, so a nearby alpha particle gets strongly repelled, while most alpha particles never come close to any nucleus and go straight. Now my fellow member will compare Thomson's and Rutherford's models.

Member 8 — Thomson vs Rutherford

MEMBER 08

Let us compare both models. The main difference is the **position of the positive charge**.

	Thomson model	Rutherford model
Positive charge	Spread throughout the atom	Concentrated in the nucleus
Electrons	Embedded in the positive sphere	Revolve around the nucleus
Nucleus	No nucleus	Nucleus present
Alpha scattering	Failed to explain it	Successfully explained it

Member 8 — Thomson vs Rutherford

MEMBER 08

So Thomson imagined a uniform, filled atom, while Rutherford showed that the atom is mostly empty with a tiny dense centre. Thomson's contribution was proving that electrons exist inside atoms. Rutherford's contribution was discovering the nucleus. Each model improved on the previous one using experimental evidence.

But there is one thing common to both: they were **classical models**, based on classical physics. And classical models also had serious limitations, which my fellow member will explain.

Member 9 — Limitations of Classical Models

MEMBER 09

Rutherford's model was a big step, but it also failed, because of classical physics.

In a circular orbit, the electron is constantly changing direction, so it is **accelerated**. Classical physics, specifically Maxwell's electromagnetic theory, says that an **accelerated charge radiates energy** in the form of electromagnetic waves. So the electron should **lose energy continuously**. As it loses energy, its orbit should shrink, and it should **spiral into the nucleus**. Calculations show this would take only about 10^{-11} second. So the **atom should collapse**. But in reality, **atoms are stable**.

Member 9 — Limitations of Classical Models

MEMBER 09

The second problem is the spectrum. Since the electron would spiral inward, its frequency would change continuously, so the classical model predicts a **continuous spectrum**, with all wavelengths. But real atoms, like hydrogen, show a **line spectrum**, only specific wavelengths such as the Balmer series in the visible region. These lines follow the Rydberg formula, which Rutherford's model could not explain.

Therefore, **classical physics fails at the atomic scale**, and a new theory was needed. Let us see how it was solved.

Member 10 — Conclusion and Modern Atomic Theory

MEMBER 10

To conclude: **Thomson** discovered the electron and proposed the plum pudding model. **Rutherford** discovered the nucleus and proposed the planetary model. But **classical models failed to explain stability and spectra**.

Member 10 — Conclusion and Modern Atomic Theory

MEMBER 10

In **1913**, Niels **Bohr** fixed this with three ideas. First, electrons move only in certain **stationary orbits** and do not radiate energy while in them. Second, only orbits where the angular momentum is a whole-number multiple of $h/2\pi$ are allowed, which means orbits are **quantised**. Third, an electron emits or absorbs light only when it jumps between orbits, and the energy is $h\nu$, the difference between the two energy levels. This explained the hydrogen line spectrum. For hydrogen, the energy levels are $-13.6/n^2$ electron volts.

Member 10 — Conclusion and Modern Atomic Theory

MEMBER 10

But Bohr's model worked only for hydrogen-like atoms. So the **modern quantum model** was developed, based on Heisenberg's uncertainty principle and Schrödinger's wave equation. It says we cannot know the exact path of an electron, only the **probability** of finding it in a region of space. Such a region is called an **orbital**, described by quantum numbers.

The timeline is: **Dalton → Thomson → Rutherford → Bohr → Quantum Model.** Atomic theory developed through experiments and the correction of old models.

Quick Revision (Any Member)

ANY MEMBER

Quick revision. The electron was discovered by J.J. Thomson in 1897. The Thomson model is the plum pudding model and has no nucleus. Rutherford's experiment is the gold foil, or alpha scattering, experiment. Rutherford discovered the nucleus, and his model is the planetary model. Classical models failed to explain stability and line spectra. Bohr introduced quantised orbits, and the modern model uses orbitals.

Closing (All Members Together)

ALL MEMBERS

Respected teacher and friends, thank you for listening to our presentation. We are ready for questions.

Teacher Cross Questions — Detailed Answers

ALL MEMBERS

Suggestion: let the member who presented that topic answer, instead of leaving all ten to Member 10.

Teacher Cross Questions — Detailed Answers

MEMBER 01

1. **What is an atom?** (Member 1) The smallest particle of an element that can take part in chemical reactions. It was once thought indivisible, but it contains electrons, protons, and neutrons.

Teacher Cross Questions — Detailed Answers

MEMBER 03

2. **Why is the Thomson model called the plum pudding model?** (Member 3)
Because electrons are embedded in a positive sphere, like plums in a pudding or seeds in a watermelon.

Teacher Cross Questions — Detailed Answers

MEMBER 02

3. **What are cathode rays?** (Member 2) Streams of negatively charged particles, called electrons, that travel from cathode to anode in a low-pressure gas tube. They are deflected by electric and magnetic fields.

Teacher Cross Questions — Detailed Answers

MEMBER 04

4. **Why did the Thomson model fail?** (Member 4) It could not explain large-angle scattering of alpha particles, had no nucleus, and could not explain line spectra or stability.

Teacher Cross Questions — Detailed Answers

MEMBER 05

5. **What are alpha particles?** (Member 5) Helium nuclei, two protons and two neutrons, with charge +2e. They are emitted by radioactive elements like radium and polonium.

Teacher Cross Questions — Detailed Answers

MEMBER 06

6. **Why did most alpha particles pass straight?** (Member 6) Because the atom is mostly empty space, and most particles never came near a nucleus.

Teacher Cross Questions — Detailed Answers

MEMBER 06

7. **Why were some alpha particles deflected greatly?** (Member 6) Because they came very close to the tiny, dense, positively charged nucleus and were strongly repelled. A small impact parameter gives a large deflection.

Teacher Cross Questions — Detailed Answers

MEMBER 07

8. **What is the Rutherford nuclear model?** (Member 7) The atom has a small, dense, positive nucleus containing most of the mass, with electrons revolving around it. Most of the atom is empty space.

Teacher Cross Questions — Detailed Answers

MEMBER 07

9. **Why is the Rutherford model called the planetary model?** (Member 7)
Because electrons revolve around the nucleus like planets around the sun, with electrostatic force playing the role of gravity.

Teacher Cross Questions — Detailed Answers

MEMBER 09

10. **What is the main limitation of classical models?** (Member 9) They cannot explain atomic stability, since an orbiting electron should radiate energy and spiral into the nucleus, and they cannot explain line spectra, since they predict a continuous one.